CSE327: Software Engineering [Spring 2026]

Term Project: Election Management System

Course: Software Engineering
Project Type: Group
Submission Format: Two Milestones (SRS + Design, then Final Report)
Goal: Apply requirements engineering, system modeling, documentation, and software design processes in the context of a large-scale public administration system.

1. Project Overview

National elections involve complex administrative processes, including voter verification, polling station management, vote recording, postal ballot handling, and the consolidation of final results.

Managing these operations requires a structured information system capable of supporting multiple election officials and stakeholders.

This project requires students to analyze, document, and design an Election Management System (EMS) that supports election operations in a simulated national election environment.

The project context is inspired by elections conducted under the Bangladesh Election Commission.

Students are expected to produce professional-level software engineering documentation including:

- A complete Software Requirements Specification (SRS)
- System modeling diagrams
- A final system design and analysis report
- A prototype demonstrating core workflows

The emphasis of the project is on requirements engineering, system modeling, architecture design, documentation quality, and analytical thinking.

2. System Description (High-Level)

The Election Management System should support the following conceptual capabilities:

Voter Services

- Search polling station using voter ID or national ID
- View assigned polling center and booth
- View election results once officially published

Polling Station Operations

- Record vote counts for each candidate
- Validate total ballots vs votes
- Submit polling station results

Election Administration

- Manage constituencies, polling stations, and booths
- Manage candidate and political party information
- Manage election officers

Result Management

- Aggregate polling station results
- Generate constituency-level results
- Identify winning candidates

Students must refine detailed requirements during SRS preparation.

3. Constraints and Assumptions

The system should consider the following conceptual constraints:

Polling Structure

- An election hierarchy exists:
- Constituency → Polling Station → Polling Booth
- Each booth belongs to a single polling station.

Voting Constraints

- Each voter may vote only once in a given election.
- Total votes cannot exceed issued ballots.
- Each candidate belongs to one constituency.

Officer Roles

Different election officials perform different tasks:

- Assistant Presiding Officer → enters vote counts
- Presiding Officer → verifies station results
- Assistant Returning Officer → compiles results
- Returning Officer → approves constituency results

The system must support searching and reviewing election data from multiple perspectives. Students must document additional assumptions where necessary.

4. Required Deliverables

This project has **two submissions**.

<u>Submission 1 (Milestone 1)</u>

SRS + System Design Models

A. Software Requirements Specification (SRS)

The SRS must include (IEEE-style structure recommended):

1. **Introduction**
   - Purpose
   - Scope
   - Definitions
   - Overview
2. **Overall Description**
   - Product perspective
   - Product functions (high-level descriptions only)
   - User characteristics
   - Constraints
   - Assumptions & dependencies
3. **System Requirements**
   - **Description of requirements (Functional and Non-Functional)**
   - Data descriptions
   - Interface descriptions
   - System rules and election constraints
   - User role permissions
4. **External Interface Descriptions**
   - UI expectations (conceptual)
   - System interaction points

B. Modeling & Design Diagrams

Students must submit the following:

- **Use Case Diagram**
- **At least 5 Use Case Descriptions**
    - Suggested use cases: Search voter information with their polling station information, Record polling station votes, Verify polling station result, Consolidate constituency result, View election results
- **Sequence Diagrams** (minimum two major interactions)
- **Activity Diagrams** (minimum two)
- **Class Diagram**
- **High-Level Architecture Diagram**
- **UI Wireframes / Mockups**
    - Students must design conceptual screens for: Voter search interface, Polling station vote entry, Result verification screen, Constituency result dashboard

Evaluation for Submission 1

Component	Weight
SRS Document	40%
Modeling Diagrams	40%
Completeness & clarity	20%

Submission 2 (Milestone 2)

Final Report + Revised Documentation + Prototype

A. Revised SRS (if needed)

Update based on feedback from Submission 1.

B. Updated Modeling & Design Documents

Incorporate instructor feedback.

C. Prototype Demonstration (Required)

A **simple prototype** must be created:

The prototype should demonstrate workflows such as:

- Searching for a voter
- Entering polling station vote counts
- Detecting vote inconsistencies
- Viewing constituency results

<u>Full software implementation is NOT required, but will be highly appreciated.</u>

D. Testing & Validation

Students must submit:

- Test plan
- At least 5 test cases
- Expected vs actual results from prototype

Validation scenarios (e.g., Duplicate vote attempt, Incorrect ballot count, Unauthorized result modification)

E. Final Report

The final report must include:

1. Title page
2. Abstract
3. Introduction
4. Summary of SRS
5. System modeling summary
6. Election management constraint analysis
7. Prototype demonstration
8. Testing summary
9. Challenges
10. Conclusion

Evaluation for Submission 2

Component	Weight
Revised SRS & Models	20%
Prototype	30%
Testing & Validation	20%
Report Quality	30%

5. Key Project Features Students Must Capture (Conceptual)

Students must clearly model and document the following system behaviors.

Voting Validation Rules

- Each voter can vote only once.
- Total votes cannot exceed issued ballots.
- Candidate votes must sum to total votes.

Result Aggregation Rules

- Polling station result submission
- Constituency-level aggregation
- Winner determination

System Views

Students must document the conceptual behavior of:

- Viewing results by constituency
- Searching voters by ID
- Viewing polling station summaries
- Monitoring election progress

6. Optional Enhancements (Bonus)

Students may conceptualize additional features for extra marks:

- Real-time election result dashboard
- Graphical vote statistics
- Fraud detection alerts

- Election monitoring analytics
- Authentication and role-based access control

7. Summary of Milestones

Milestone	**Deliverables**	**Due**
Submission 1	SRS + Modeling Diagrams	13 April 2026 (Monday)
Submission 2	Final Report + Prototype + Updated Models	TBA
Final Presentation	A brief Demonstration + Q/A	TBA